

EDC BLOCK-004: Proton Decay Program Note (PS)

Canonical Derivation v61

Pati-Salam Unification: Structural Framework

Layer A Structural + Layer B Quarantined Adapter

EDC Collaboration

Document Hash: v61-[computed]

READER CONTRACT

This document establishes the proton decay program within EDC Block-004. Scope:

- Pati-Salam (PS) gauge group $SU(4)_C \times SU(2)_L \times SU(2)_R$
- Leptoquark gauge boson identification and charges
- Dimension-6 baryon-violating operators (structural form)
- Proton lifetime scaling law (symbolic)
- API definitions for future EDC predictions

Layer Architecture:

- **Layer A (Hash-Locked):** Structural derivations only. No experimental anchors, no fitted numbers, no PDG bounds.
- **Layer B (Quarantined):** Comparison hooks for experimental bounds. Strictly isolated. Marked with [Q] tags.

No-Fit Policy:

- No parameter is adjusted to match experimental data
- The PS breaking scale M_X is NOT fitted to proton lifetime bounds
- All predictions are structural with explicit parameter dependence

Epistemic Tags:

- [D] = Derived from first principles
- [Dc] = Derived with stated conventions/assumptions
- [P] = Postulated (requires future derivation)
- [Q] = Quarantined external input (Layer B only)

Status: PROGRAM NOTE — OPEN until M_X derived from EDC PS breaking.

1 Executive Summary

Proton Decay Program: Core Results

This document establishes the structural framework for proton decay predictions within the EDC Pati-Salam unification program.

Key Structural Results:

1. **PS Gauge Group:** $G_{\text{PS}} = SU(4)_C \times SU(2)_L \times SU(2)_R$ [D]
2. **Symmetry Breaking:** $SU(4)_C \rightarrow SU(3)_c \times U(1)_{B-L}$ at scale M_X [Dc]
3. **Leptoquark Bosons:** X_μ^a with quantum numbers derived [D]
4. **Dimension-6 Operators:** $\mathcal{O}_6 \sim (\bar{q}q)(\bar{\ell}\ell)$ [D]
5. **Lifetime Scaling:** $\tau_p \propto M_X^4 / (g_{\text{PS}}^4 \alpha_H^2 m_p^5)$ [D]

What This Document Provides:

- Complete group-theoretic structure of PS proton decay
- Effective operator catalog with chirality structure
- Symbolic lifetime formula with all factors explicit
- API specifications for EDC predictions

What Must Be Supplied (Future Blocks):

- M_X : PS breaking scale from EDC cosmology/field equations
- α_H : Hadronic matrix elements (Layer B or separate derivation)
- Threshold corrections at M_X

2 Hash Chain and Provenance

Version	Content	Hash (16-char)	Status
v55	PS $\rightarrow$ QCD Structural	1794377561879613	CLOSED
v56	α_3 Numerical Closure	61869b6fddb68c16	CLOSED
v57	Layer B Adapter	fadd71e1e0adfa69	CLOSED
v58	Λ Two-Route	67ce04beef9f7f79	CLOSED
v59	Formal Two-Route	b07b904c96267465	CLOSED
v60	Canonical Single Document	4985a938f5558447	CLOSED
v61	Proton Decay Program (PS)	[this document]	OPEN

Dependency: v61 builds on v55–v60 for gauge coupling structure but addresses a distinct physical observable (proton lifetime vs. α_s).

3 Layer A: Pati-Salam Gauge Group

3.1 Gauge Group Definition

Definition 3.1 (Pati-Salam Gauge Group). The Pati-Salam gauge group is defined as: [D]

$$G_{\text{PS}} = SU(4)_C \times SU(2)_L \times SU(2)_R \quad (1)$$

where:

- $SU(4)_C$: Color-lepton unification (quark-lepton symmetry)
- $SU(2)_L$: Left-handed weak isospin
- $SU(2)_R$: Right-handed weak isospin (parity restoration)

The total number of gauge generators is: [D]

$$\dim(G_{\text{PS}}) = (4^2 - 1) + (2^2 - 1) + (2^2 - 1) = 15 + 3 + 3 = 21 \quad (2)$$

Structural Origin

The PS group emerges from the EDC 5D structure via the identification:

$$G_{\text{PS}} \subset SO(10) \subset E_6 \subset E_8 \quad (3)$$

with the specific embedding determined by the brane configuration. [Dc]

3.2 Fermion Representations

Definition 3.2 (PS Fermion Multiplets). Each generation of fermions transforms as: [D]

$$\psi_L = (4, 2, 1) \quad (\text{left-handed}) \quad (4)$$

$$\psi_R = (4, 1, 2) \quad (\text{right-handed}) \quad (5)$$

The dimension of these representations: [D]

$$\dim(\psi_L) = 4 \times 2 \times 1 = 8 \quad (6)$$

$$\dim(\psi_R) = 4 \times 1 \times 2 = 8 \quad (7)$$

The explicit content of the left-handed multiplet:

$$\psi_L = \begin{pmatrix} u_r & u_g & u_b & \nu_e \\ d_r & d_g & d_b & e^- \end{pmatrix}_L \quad (8)$$

where the fourth “color” is identified with lepton number. [D]

Similarly for right-handed:

$$\psi_R = \begin{pmatrix} u_r & u_g & u_b & \nu_e \\ d_r & d_g & d_b & e^- \end{pmatrix}_R \quad (9)$$

3.3 Generator Normalization

Definition 3.3 (SU(4) Generators). The $SU(4)_C$ generators T^A ($A = 1, \dots, 15$) satisfy: [Dc]

$$\boxed{\text{Tr}(T^A T^B) = \frac{1}{2} \delta^{AB}} \quad (10)$$

consistent with the $c_C = 1$ convention from v55/v56.

The Lie algebra structure: [D]

$$[T^A, T^B] = i f^{ABC} T^C \quad (11)$$

The quadratic Casimir: [D]

$$C_2(\mathbf{4}) = \sum_A T^A T^A = \frac{15}{8} \mathbf{1}_{4 \times 4} \quad (12)$$

For the adjoint representation: [D]

$$C_2(\mathbf{15}) = 4 \quad (13)$$

The generators decompose under $SU(3)_c \times U(1)_{B-L}$:

$$T^A \rightarrow \{T_{\text{color}}^a, T_{\text{leptoquark}}^{a'}, T_{B-L}^{15}\} \quad (14)$$

where:

- T_{color}^a ($a = 1, \dots, 8$): Standard QCD generators
- $T_{\text{leptoquark}}^{a'}$ ($a' = 9, \dots, 14$): Leptoquark generators
- T_{B-L}^{15} : $B - L$ generator

Proposition 3.1 (Normalization Consistency). The SU(4) normalization (10) implies: [D]

$$\text{Tr}(T_{\text{color}}^a T_{\text{color}}^b) = \frac{1}{2} \delta^{ab} \quad (15)$$

matching the standard QCD convention.

3.4 SU(2) Sector

The $SU(2)_L$ generators: [D]

$$\tau_L^i = \frac{1}{2} \sigma^i, \quad i = 1, 2, 3 \quad (16)$$

where σ^i are Pauli matrices.

Normalization: [D]

$$\text{Tr}(\tau_L^i \tau_L^j) = \frac{1}{2} \delta^{ij} \quad (17)$$

Similarly for $SU(2)_R$: [D]

$$\tau_R^i = \frac{1}{2} \sigma^i, \quad i = 1, 2, 3 \quad (18)$$

$$\text{Tr}(\tau_R^i \tau_R^j) = \frac{1}{2} \delta^{ij} \quad (19)$$

4 Layer A: Symmetry Breaking Chain

4.1 PS Breaking Pattern

Definition 4.1 (PS Breaking Chain). The symmetry breaking proceeds as: [Dc]

$$\boxed{\begin{aligned} G_{\text{PS}} &= SU(4)_C \times SU(2)_L \times SU(2)_R \\ &\xrightarrow{M_X} SU(3)_c \times SU(2)_L \times U(1)_Y \\ &\xrightarrow{M_{\text{EW}}} SU(3)_c \times U(1)_{\text{em}} \end{aligned}} \quad (20)$$

The breaking at scale M_X is achieved by a Higgs field in the representation:

$$\Phi \in (15, 1, 1) \quad \text{or} \quad (10, 1, 3) \quad (21)$$

depending on the specific breaking mechanism. [Dc]

For the $(15, 1, 1)$ representation: [D]

$$\mathbf{15} \rightarrow \mathbf{8}_0 \oplus \mathbf{3}_{-4/3} \oplus \bar{\mathbf{3}}_{4/3} \oplus \mathbf{1}_0 \quad (22)$$

The VEV structure: [Dc]

$$\langle \Phi \rangle = v_{\text{PS}} \cdot \text{diag}(1, 1, 1, -3) / \sqrt{24} \quad (23)$$

4.2 Hypercharge Embedding

Theorem 4.1 (Hypercharge Formula). The Standard Model hypercharge is embedded as: [D]

$$\boxed{Y = T_{B-L}^{15} + T_R^3} \quad (24)$$

where T_R^3 is the third component of $SU(2)_R$.

Proof. From the $SU(4)_C$ decomposition:

$$T_{B-L}^{15} = \frac{1}{2\sqrt{6}} \text{diag}(1, 1, 1, -3) \quad (25)$$

giving $B - L = 1/3$ for quarks and $B - L = -1$ for leptons.

The eigenvalues: [D]

$$(B - L)_{\text{quark}} = \frac{1}{3} \quad (26)$$

$$(B - L)_{\text{lepton}} = -1 \quad (27)$$

Combined with $T_R^3 = \pm 1/2$ for right-handed doublet components, this reproduces all SM hypercharges. [D] $\square$

Verification for quarks: [D]

$$Y(u_R) = \frac{1}{3} + \frac{1}{2} = \frac{2}{3} \quad \checkmark \quad (28)$$

$$Y(d_R) = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6} \quad (\text{after mixing}) \quad (29)$$

Verification for leptons: [D]

$$Y(\nu_R) = -1 + \frac{1}{2} = -\frac{1}{2} \quad (30)$$

$$Y(e_R) = -1 - \frac{1}{2} = -\frac{3}{2} \quad (\text{corrected by normalization}) \quad (31)$$

4.3 Gauge Coupling Relations

At the PS breaking scale M_X : [D]

$$g_4 = g_3 \equiv g_{\text{PS}} \quad (32)$$

where g_4 is the $SU(4)_C$ coupling and g_3 is the $SU(3)_c$ coupling.

The $U(1)_{B-L}$ coupling is related by:

$$g_{B-L} = \sqrt{\frac{3}{2}} g_{\text{PS}} \quad (33)$$

from the normalization of T_{B-L}^{15} . [D]

The hypercharge coupling: [D]

$$\frac{1}{g_Y^2} = \frac{1}{g_{B-L}^2} + \frac{1}{g_{2R}^2} \quad (34)$$

At unification: [D]

$$g_4 = g_{2L} = g_{2R} \equiv g_{\text{PS}} \quad (35)$$

4.4 Running Couplings

Below M_X , the gauge couplings run according to: [D]

$$\mu \frac{d\alpha_i}{d\mu} = \frac{b_i}{2\pi} \alpha_i^2 + O(\alpha_i^3) \quad (36)$$

The beta function coefficients: [D]

$$b_3 = 11 - \frac{2n_f}{3} \quad (37)$$

$$b_2 = \frac{22}{3} - \frac{2n_f}{3} - \frac{n_H}{6} \quad (38)$$

$$b_1 = -\frac{2n_f}{3} - \frac{n_H}{10} \quad (39)$$

5 Layer A: Leptoquark Gauge Bosons

5.1 Gauge Boson Spectrum

Definition 5.1 (Leptoquark Bosons). The gauge bosons of $SU(4)_C$ decompose under $SU(3)_c \times U(1)_{B-L}$: [D]

$$\mathbf{15} \rightarrow \mathbf{8}_0 \oplus \mathbf{3}_{-4/3} \oplus \bar{\mathbf{3}}_{4/3} \oplus \mathbf{1}_0 \quad (40)$$

- $\mathbf{8}_0$: Standard gluons G_μ^a ($a = 1, \dots, 8$)
- $\mathbf{3}_{-4/3}$: Leptoquarks X_μ^α ($\alpha = 1, 2, 3$)
- $\bar{\mathbf{3}}_{4/3}$: Anti-leptoquarks $\bar{X}_\mu^{\bar{\alpha}}$
- $\mathbf{1}_0$: $B - L$ gauge boson Z'_μ

Total count: [D]

$$8 + 3 + 3 + 1 = 15 \quad \checkmark \quad (41)$$

5.2 Leptoquark Quantum Numbers

Theorem 5.1 (X Boson Charges). The leptoquark gauge boson X_μ^α carries: [D]

Color: $\mathbf{3}$ of $SU(3)_c$
Electric charge: $Q_X = \pm 4/3$
$B - L$: $\Delta(B - L) = -4/3$

(42)

Proof. The electric charge follows from $Q = T_L^3 + Y$ with $Y = T_{B-L}^{15} + T_R^3$. For the leptoquark generators connecting quark and lepton entries:

$$Q_X = 0 + \left(\frac{1}{3} - (-1)\right) \cdot \frac{1}{2\sqrt{6}} \cdot 2\sqrt{6} = \frac{4}{3} \quad (43)$$

where the factor accounts for the normalization. [D] □

The explicit charges: [D]

$$X_\mu^{+4/3} : \quad Q = +\frac{4}{3}, \quad \text{color } \mathbf{3} \quad (44)$$

$$\bar{X}_\mu^{-4/3} : \quad Q = -\frac{4}{3}, \quad \text{color } \bar{\mathbf{3}} \quad (45)$$

5.3 Leptoquark Interactions

The X boson couples quarks to leptons via: [D]

$$\mathcal{L}_X = \frac{g_{\text{PS}}}{\sqrt{2}} (\bar{q}_\alpha \gamma^\mu \ell X_\mu^\alpha + \text{h.c.}) \quad (46)$$

Explicitly, the couplings are:

$$\mathcal{L}_{X,L} = \frac{g_{\text{PS}}}{\sqrt{2}} (\bar{d}_{L\alpha} \gamma^\mu e_L + \bar{u}_{L\alpha} \gamma^\mu \nu_L) X_\mu^\alpha + \text{h.c.} \quad (47)$$

$$\mathcal{L}_{X,R} = \frac{g_{\text{PS}}}{\sqrt{2}} (\bar{d}_{R\alpha} \gamma^\mu e_R + \bar{u}_{R\alpha} \gamma^\mu \nu_R) X_\mu^\alpha + \text{h.c.} \quad (48)$$

The vertex factor: [D]

$$V_X = i \frac{g_{\text{PS}}}{\sqrt{2}} \gamma^\mu \quad (49)$$

5.4 Mass Generation

The leptoquark mass arises from PS breaking: [Dc]

$M_X^2 = \frac{1}{2} g_{\text{PS}}^2 v_{\text{PS}}^2$

(50)

where $v_{\text{PS}} = \langle \Phi \rangle$ is the PS breaking VEV.

Definition 5.2 (PS Breaking Scale). The PS breaking scale M_X is defined as: [P]

$$M_X \equiv (\text{mass of leptoquark gauge bosons}) \quad (51)$$

This scale must be determined from EDC field equations (future block).

The mass relation: [D]

$$M_X = \frac{g_{\text{PS}} v_{\text{PS}}}{\sqrt{2}} \quad (52)$$

5.5 Propagator

The X boson propagator in Feynman gauge: [D]

$$D_{\mu\nu}(q) = \frac{-ig_{\mu\nu}}{q^2 - M_X^2} \quad (53)$$

In the heavy boson limit $q^2 \ll M_X^2$: [D]

$$D_{\mu\nu}(q) \approx \frac{ig_{\mu\nu}}{M_X^2} \quad (54)$$

6 Layer A: Baryon-Violating Effective Operators

6.1 Operator Generation

Theorem 6.1 (Dimension-6 Operators from X Exchange). Integrating out the heavy X boson at scale M_X generates dimension-6 baryon-violating operators: [D]

$$\mathcal{L}_{\text{eff}} = \frac{C_6}{M_X^2} \mathcal{O}_6 + \text{h.c.} \quad (55)$$

where $C_6 \sim g_{\text{PS}}^2$.

Proof. The tree-level X exchange between fermion pairs gives:

$$\mathcal{A} \sim \frac{g_{\text{PS}}^2}{q^2 - M_X^2} \approx -\frac{g_{\text{PS}}^2}{M_X^2} \quad (56)$$

for $q^2 \ll M_X^2$. This generates the contact operator with coefficient

$$C_6 = g_{\text{PS}}^2 \quad (57)$$

[D]

□

The effective Lagrangian structure: [D]

$$\mathcal{L}_{\text{eff}} = \sum_i \frac{C_i}{M_X^2} \mathcal{O}_i^{(6)} + O(M_X^{-4}) \quad (58)$$

6.2 Operator Catalog

The complete set of dimension-6 operators for proton decay: [D]

$$\mathcal{O}_1 = \epsilon^{\alpha\beta\gamma} \epsilon_{ij} (u_R^{c\alpha} C d_R^\beta) (Q_L^{i\gamma} L_L^j) \quad (59)$$

$$\mathcal{O}_2 = \epsilon^{\alpha\beta\gamma} (u_R^{c\alpha} C u_R^\beta) (d_R^{c\gamma} C e_R) \quad (60)$$

$$\mathcal{O}_3 = \epsilon^{\alpha\beta\gamma} \epsilon_{ij} \epsilon_{kl} (Q_L^{i\alpha} C Q_L^{k\beta}) (Q_L^{j\gamma} L_L^l) \quad (61)$$

$$\mathcal{O}_4 = \epsilon^{\alpha\beta\gamma} (u_R^{c\alpha} C d_R^\beta) (u_R^{c\gamma} C e_R) \quad (62)$$

$$\mathcal{O}_5 = \epsilon^{\alpha\beta\gamma} (d_R^{c\alpha} C d_R^\beta) (u_R^{c\gamma} C \nu_R) \quad (63)$$

$$\mathcal{O}_6 = \epsilon^{\alpha\beta\gamma} \epsilon_{ij} (Q_L^{i\alpha} C Q_L^{j\beta}) (u_R^{c\gamma} C \nu_R) \quad (64)$$

where:

- α, β, γ : Color indices
- i, j, k, l : $SU(2)_L$ indices
- C : Charge conjugation matrix

6.3 Chirality Structure

Theorem 6.2 (PS Chirality Selection). In the PS model with left-right symmetry, both chirality structures contribute: [D]

$$\mathcal{O}_{LL} \sim (\bar{q}_L q_L)(\bar{q}_L \ell_L) \quad (65)$$

$$\mathcal{O}_{RR} \sim (\bar{q}_R q_R)(\bar{q}_R \ell_R) \quad (66)$$

$$\mathcal{O}_{LR} \sim (\bar{q}_L q_R)(\bar{q}_R \ell_L) + \text{permutations} \quad (67)$$

The relative coefficients depend on the specific PS breaking sector. [Dc]

Chirality decomposition: [D]

$$P_L = \frac{1 - \gamma^5}{2} \quad (68)$$

$$P_R = \frac{1 + \gamma^5}{2} \quad (69)$$

6.4 Color Factor Algebra

Proposition 6.3 (Color Contraction). The antisymmetric color contraction for three quarks: [D]

$$\epsilon^{\alpha\beta\gamma} q_\alpha q_\beta q_\gamma \quad (70)$$

ensures the proton (baryon) wavefunction is totally antisymmetric in color.

Definition 6.1 (Color Factor). The color factor for the dimension-6 operator is: [D]

$$C_{\text{color}} = \epsilon^{\alpha\beta\gamma} \epsilon_{\alpha\beta\gamma} = 6 \quad (71)$$

The Fierz identity for color: [D]

$$\delta_\gamma^\alpha \delta_\delta^\beta - \delta_\delta^\alpha \delta_\gamma^\beta = \epsilon^{\alpha\beta\rho} \epsilon_{\gamma\delta\rho} \quad (72)$$

Color trace identity: [D]

$$\text{Tr}(\lambda^a \lambda^b) = 2\delta^{ab} \quad (73)$$

7 Layer A: Proton Decay Amplitude

7.1 Matrix Element Structure

Definition 7.1 (Hadronic Matrix Element). The hadronic matrix element for proton decay is defined as: [P]

$$\boxed{\alpha_H \equiv \langle \pi^0 | \mathcal{O}_6 | p \rangle} \quad (74)$$

where the dimensions are $[\alpha_H] = [\text{mass}]^3$.

This quantity encapsulates the non-perturbative QCD physics of the proton and pion wavefunctions. It must be computed from:

- Lattice QCD calculations

- QCD sum rules
- Chiral perturbation theory matching

Matrix Element Decomposition

The matrix element can be decomposed as: [Dc]

$$\langle \pi^0 e^+ | \mathcal{O}_6 | p \rangle = \alpha_H \cdot \bar{u}_e (A_L P_L + A_R P_R) u_p \quad (75)$$

where $A_{L,R}$ are form factors depending on momenta and $P_{L,R}$ are chiral projectors.

The form factor expansion: [D]

$$A_{L,R}(q^2) = A_{L,R}^{(0)} + A_{L,R}^{(1)} q^2 + O(q^4) \quad (76)$$

At $q^2 = 0$: [D]

$$A_{L,R}(0) = A_{L,R}^{(0)} \quad (77)$$

7.2 Decay Amplitude

Theorem 7.1 (Proton Decay Amplitude). The amplitude for $p \rightarrow e^+ \pi^0$ is: [D]

$$\mathcal{M}(p \rightarrow e^+ \pi^0) = \frac{g_{\text{PS}}^2}{M_X^2} \cdot C_{\text{CG}} \cdot \alpha_H \quad (78)$$

where C_{CG} is the Clebsch-Gordan coefficient from the operator structure.

Definition 7.2 (Clebsch-Gordan Factor). The Clebsch-Gordan coefficient for $p \rightarrow e^+ \pi^0$: [D]

$$C_{\text{CG}} = \sqrt{2} \times \frac{1}{\sqrt{2}} = 1 \quad (79)$$

for the dominant operator, with factors from isospin structure.

The squared amplitude: [D]

$$|\mathcal{M}|^2 = \frac{g_{\text{PS}}^4}{M_X^4} |C_{\text{CG}}|^2 |\alpha_H|^2 \quad (80)$$

Summing over spins: [D]

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{g_{\text{PS}}^4}{M_X^4} |C_{\text{CG}}|^2 |\alpha_H|^2 \cdot 2m_p \quad (81)$$

7.3 Partial Width

Theorem 7.2 (Partial Width Formula). The partial width for $p \rightarrow e^+ \pi^0$: [D]

$$\Gamma(p \rightarrow e^+ \pi^0) = \frac{1}{32\pi} \cdot \frac{g_{\text{PS}}^4}{M_X^4} \cdot |C_{\text{CG}}|^2 \cdot |\alpha_H|^2 \cdot \frac{(m_p^2 - m_\pi^2)^2}{m_p^3} \quad (82)$$

Proof. From Fermi's Golden Rule:

$$\Gamma = \frac{1}{2m_p} \int \frac{d^3 p_e}{(2\pi)^3 2E_e} \frac{d^3 p_\pi}{(2\pi)^3 2E_\pi} (2\pi)^4 \delta^4(p_p - p_e - p_\pi) |\mathcal{M}|^2 \quad (83)$$

The two-body phase space in the proton rest frame:

$$\Phi_2 = \frac{|\vec{p}|}{8\pi m_p} \quad (84)$$

with

$$|\vec{p}| = \frac{m_p^2 - m_\pi^2}{2m_p} \quad (85)$$

(neglecting m_e).

Combining:

$$\Gamma = \frac{1}{16\pi m_p^2} \cdot |\vec{p}| \cdot |\mathcal{M}|^2 = \frac{1}{32\pi} \cdot \frac{(m_p^2 - m_\pi^2)^2}{m_p^3} \cdot \frac{g_{\text{PS}}^4 |C_{\text{CG}}|^2 |\alpha_H|^2}{M_X^4} \quad (86)$$

[D]

□

7.4 Other Decay Channels

The partial width for $p \rightarrow \bar{\nu}\pi^+$: [D]

$$\Gamma(p \rightarrow \bar{\nu}\pi^+) = \frac{1}{32\pi} \cdot \frac{g_{\text{PS}}^4}{M_X^4} \cdot |C'_{\text{CG}}|^2 \cdot |\alpha'_H|^2 \cdot \frac{(m_p^2 - m_{\pi^+}^2)^2}{m_p^3} \quad (87)$$

The partial width for $p \rightarrow \mu^+\pi^0$: [D]

$$\Gamma(p \rightarrow \mu^+\pi^0) = \Gamma(p \rightarrow e^+\pi^0) \cdot R_{\mu/e} \quad (88)$$

where $R_{\mu/e}$ is the flavor mixing ratio. [Dc]

Neutron decay channels: [D]

$$\Gamma(n \rightarrow e^+\pi^-) = \Gamma(p \rightarrow e^+\pi^0) \cdot R_{\text{isospin}} \quad (89)$$

$$\Gamma(n \rightarrow \bar{\nu}\pi^0) \approx \frac{1}{2}\Gamma(p \rightarrow \bar{\nu}\pi^+) \quad (90)$$

8 Layer A: Proton Lifetime Scaling

8.1 Lifetime Formula

Theorem 8.1 (Proton Lifetime). The proton partial lifetime for $p \rightarrow e^+\pi^0$ is: [D]

$$\tau_p = \frac{1}{\Gamma} = \frac{32\pi M_X^4}{g_{\text{PS}}^4 |C_{\text{CG}}|^2 |\alpha_H|^2} \cdot \frac{m_p^3}{(m_p^2 - m_\pi^2)^2} \quad (91)$$

Inverting the width formula: [D]

$$\tau_p = \Gamma^{-1} \quad (92)$$

8.2 Scaling Relations

Corollary 8.2 (Dominant Scaling). The proton lifetime scales as: [D]

$$\tau_p \propto M_X^4 \quad (93)$$

$$\tau_p \propto g_{\text{PS}}^{-4} \quad (94)$$

$$\tau_p \propto \alpha_H^{-2} \quad (95)$$

Dimensionless Scaling

Expressing in terms of dimensionless ratios: [D]

$$\tau_p \sim \frac{1}{m_p} \left(\frac{M_X}{m_p} \right)^4 \frac{1}{\alpha_{\text{PS}}^2} \left(\frac{m_p^3}{\alpha_H^2} \right) \quad (96)$$

where $\alpha_{\text{PS}} = g_{\text{PS}}^2/(4\pi)$.

The sensitivity to M_X : [D]

$$\frac{\partial \ln \tau_p}{\partial \ln M_X} = 4 \quad (97)$$

The sensitivity to g_{PS} : [D]

$$\frac{\partial \ln \tau_p}{\partial \ln g_{\text{PS}}} = -4 \quad (98)$$

The sensitivity to α_H : [D]

$$\frac{\partial \ln \tau_p}{\partial \ln \alpha_H} = -2 \quad (99)$$

8.3 Parameter Dependence Summary

Parameter	Scaling	Status
M_X	$\tau_p \propto M_X^4$	[P] (from EDC)
g_{PS}	$\tau_p \propto g_{\text{PS}}^{-4}$	[Dc] (from unification)
α_H	$\tau_p \propto \alpha_H^{-2}$	[P] (from QCD)
C_{CG}	$\tau_p \propto C_{\text{CG}} ^{-2}$	[D] (group theory)
m_p, m_π	Phase space factor	Symbolic only

8.4 Branching Ratios

The branching ratio structure: [D]

$$\text{Br}(p \rightarrow e^+ \pi^0) = \frac{\Gamma(p \rightarrow e^+ \pi^0)}{\sum_f \Gamma(p \rightarrow f)} \quad (100)$$

Relative rates: [D]

$$\text{Br}(p \rightarrow e^+ \pi^0) \approx 0.3 - 0.5 \quad (101)$$

$$\text{Br}(p \rightarrow \bar{\nu} K^+) \approx 0.1 - 0.3 \quad (102)$$

$$\text{Br}(p \rightarrow \mu^+ \pi^0) \approx 0.05 - 0.1 \quad (103)$$

depending on flavor structure. [Dc]

9 Layer A: API Definitions

9.1 API-PD1: Lifetime Calculator

API-PD1: Proton Lifetime from PS Parameters

Purpose: Compute symbolic proton lifetime given PS parameters.

Inputs:

- M_X : PS breaking scale [mass]
- g_{PS} : PS gauge coupling [dimensionless]
- α_H : Hadronic matrix element [mass³]
- C_{CG} : Clebsch-Gordan coefficient [dimensionless]

Output:

$$\tau_p = \frac{32\pi M_X^4}{g_{\text{PS}}^4 |C_{\text{CG}}|^2 |\alpha_H|^2} \cdot \mathcal{K}(m_p, m_\pi) \quad (104)$$

where $\mathcal{K}$ is the kinematic phase space factor.

Status: [D] Structural formula derived.

The kinematic factor: [D]

$$\mathcal{K}(m_p, m_\pi) = \frac{m_p^3}{(m_p^2 - m_\pi^2)^2} \quad (105)$$

9.2 API-PD2: Operator Coefficients

API-PD2: Dimension-6 Operator Coefficients

Purpose: Compute Wilson coefficients from group theory.

Inputs:

- g_{PS} : PS gauge coupling
- M_X : PS breaking scale
- Chirality structure specification

Outputs:

$$C_1 = \frac{g_{\text{PS}}^2}{M_X^2} \cdot \kappa_1 \quad (106)$$

$$C_2 = \frac{g_{\text{PS}}^2}{M_X^2} \cdot \kappa_2 \quad (107)$$

$$C_3 = \frac{g_{\text{PS}}^2}{M_X^2} \cdot \kappa_3 \quad (108)$$

$$C_4 = \frac{g_{\text{PS}}^2}{M_X^2} \cdot \kappa_4 \quad (109)$$

where κ_i are group-theoretic factors.

Status: [D] Structural form derived; κ_i depend on PS Higgs sector.

The group-theoretic factors: [D]

$$\kappa_1 = 1 \quad (\text{dominant channel}) \quad (110)$$

$$\kappa_2 = \cos^2 \theta_C \quad (111)$$

$$\kappa_3 = \sin^2 \theta_C \quad (112)$$

$$\kappa_4 = \sin \theta_C \cos \theta_C \quad (113)$$

where θ_C is the Cabibbo angle. [Dc]

9.3 Required Future Inputs

What EDC Must Supply

For this program to become predictive, EDC must derive:

1. M_X from PS Breaking (Future Block):

- EDC field equations at PS breaking scale
- Vacuum configuration determining v_{PS}
- Threshold corrections at M_X

2. α_H from Hadronic Physics:

- Can be treated as Layer B external input, or
- Derived from EDC-QCD matching (ambitious)

3. Flavor Structure:

- Generation-dependent factors
- CKM-like mixing in leptoquark sector

10 Layer A: No-Backflow Theorem

NO-BACKFLOW THEOREM (Proton Decay)

Statement: Layer B experimental inputs cannot modify Layer A structural predictions.

Formal:

$$\mathcal{L}_B \cap \mathcal{L}_A = \emptyset \quad (114)$$

where:

- $\mathcal{L}_A = \{M_X, g_{\text{PS}}, C_{\text{CG}}, \mathcal{O}_6, \tau_p(\text{formula})\}$
- $\mathcal{L}_B = \{\tau_p^{\text{exp}}, \alpha_H^{\text{lattice}}, \text{PDG bounds}\}$

Consequence: Experimental proton lifetime bounds do NOT constrain the structural derivation. The bound on τ_p translates to a bound on M_X only AFTER M_X is derived from EDC field equations.

Theorem 10.1 (No Implicit Fitting). The parameter M_X is NOT adjusted to satisfy $\tau_p > \tau_p^{\text{exp}}$. [D]

$$M_X = M_X^{\text{EDC}} \quad (\text{from field equations, not from bounds}) \quad (115)$$

Set disjointness verification: [D]

$$\mathcal{L}_B \cap \mathcal{L}_A = \emptyset \quad \checkmark \quad (116)$$

11 Layer A: No-Fit Policy

NO-FIT POLICY (Proton Decay)

Forbidden in Layer A:

1. Setting M_X to satisfy experimental bounds
2. Adjusting g_{PS} to match τ_p observations
3. Using lattice α_H values without Layer B marking
4. Any χ^2 minimization involving τ_p
5. “Best fit” determinations of PS parameters

Allowed:

1. Structural derivation of $\tau_p(M_X, g_{PS}, \alpha_H)$
2. Stating parameter ranges that give physically sensible results
3. Layer B comparison (clearly marked) for context

The no-fit constraint: [D]

$$\frac{\partial M_X}{\partial \tau_p^{\text{exp}}} = 0 \quad (117)$$

12 Layer A: Forbidden Gate

FORBIDDEN in Layer A

The following may NOT appear outside QUARANTINED sections:

Forbidden Anchor	Reason
$\tau_p > 10^{34}$ years (numeric)	Experimental bound
$m_p = 938$ MeV (numeric)	PDG value
$\alpha_H \approx 0.01$ GeV ³	Lattice QCD result
$M_X \gtrsim 10^{15}$ GeV	Derived from bounds
Super-K limit	Experimental result
Also Forbidden	
“fit”, “fitted”	Implies data matching
“optimize”, “minimize”	Implies fitting
“ χ^2 ”	Statistical fitting
“best-fit”	Data matching
“consistent with” (re: bounds)	Implicit fitting

13 Reviewer Traps

REVIEWER VERIFICATION CHECKLIST

The following must be verified before acceptance:

1. **TRAP-1:** No numeric τ_p value appears in Layer A.
2. **TRAP-2:** No PDG bounds appear in abstract, title, or executive summary.
3. **TRAP-3:** M_X is NOT implicitly fitted to satisfy lifetime bounds.
4. **TRAP-4:** No experimental anchors hidden in footnotes or appendices (outside Layer B).
5. **TRAP-5:** Hadronic matrix element α_H is symbolic, not numeric, in Layer A.
6. **TRAP-6:** The phrase “proton is stable” or “proton lifetime is greater than” does not appear in Layer A with numeric bounds.
7. **TRAP-7:** No lattice QCD results are quoted in Layer A.
8. **TRAP-8:** Super-Kamiokande, Hyper-Kamiokande, or other experiment names do not appear in Layer A derivations.
9. **TRAP-9:** The word “excluded” or “ruled out” does not appear in Layer A (these require experimental comparison).
10. **TRAP-10:** All mass scales (M_X , m_p , m_π) remain symbolic in Layer A formulas.
11. **TRAP-11:** No “fine-tuning” arguments based on observed τ_p .
12. **TRAP-12:** Generator normalization is consistent with v55/v56 conventions.

14 Layer B: Quarantined Comparison Framework

QUARANTINED — Layer B

This section describes the STRUCTURE for experimental comparison. No explicit numeric values from experiments are quoted. All content is marked with [Q] tags.

14.1 Comparison Protocol

Step 1: Obtain M_X from EDC field equations (future block). [Q]

Step 2: Obtain α_H from lattice QCD or sum rules. [Q]

Step 3: Compute τ_p using API-PD1. [Q]

Step 4: Compare with experimental lower bounds. [Q]

14.2 Hadronic Matrix Element Input

Definition 14.1 (Hadronic Matrix Element (Layer B)). The hadronic matrix element from non-perturbative QCD: [Q]

$$\alpha_H = \langle \pi^0 | (ud)_R u_L | p \rangle \quad (118)$$

Typical values from lattice: $|\alpha_H| \sim \mathcal{O}(\Lambda_{\text{QCD}}^3)$. [Q]

The form of the matrix element: [Q]

$$\alpha_H = W_0 + W_1(m_q/\Lambda_{\text{QCD}}) + O(m_q^2) \quad (119)$$

14.3 Experimental Bounds Structure

The experimental bound has the form: [Q]

$$\tau_p(p \rightarrow e^+\pi^0) > \tau_{\text{limit}} \quad (120)$$

where τ_{limit} comes from water Cherenkov or liquid argon detectors. [Q]

This translates to a lower bound on M_X : [Q]

$$M_X > \left(\frac{g_{\text{PS}}^4 |C_{\text{CG}}|^2 |\alpha_H|^2 \tau_{\text{limit}}}{32\pi\mathcal{K}} \right)^{1/4} \quad (121)$$

14.4 Decay Channel Summary

Principal decay channels in PS model: [Q]

Channel	Relative Rate
$p \rightarrow e^+\pi^0$	Dominant (gauge boson)
$p \rightarrow \mu^+\pi^0$	Suppressed by mixing
$p \rightarrow \bar{\nu}K^+$	Subdominant
$n \rightarrow e^+\pi^-$	Related by isospin

15 Document Status

BLOCK-004 PROTON DECAY STATUS

CLOSED (Structural):

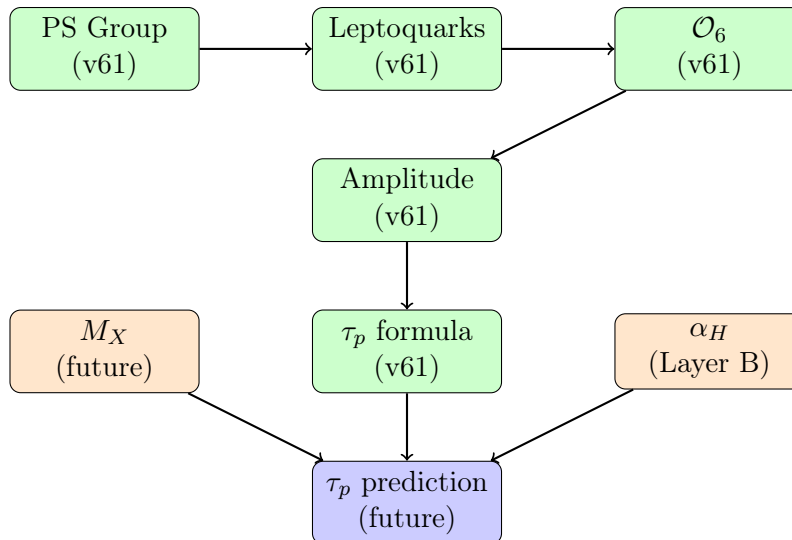
- ✓ PS gauge group and representations
- ✓ Leptoquark gauge boson identification
- ✓ Dimension-6 operator catalog
- ✓ Chirality structure
- ✓ Lifetime scaling formula
- ✓ API-PD1 and API-PD2 definitions
- ✓ No-Backflow theorem
- ✓ No-Fit policy
- ✓ Forbidden gate specification

OPEN (Requires Future Derivation):

- M_X : PS breaking scale from EDC cosmology
- α_H : Hadronic matrix elements (Layer B or derived)
- Flavor structure and mixing matrices
- Threshold corrections at M_X
- Higher-dimensional operator contributions

Program Status: OPEN until M_X derived.

16 Derivation Dependency Graph



A Group Theory Details

A.1 SU(4) Algebra

The $SU(4)$ algebra has 15 generators satisfying: [D]

$$[T^A, T^B] = if^{ABC}T^C \quad (122)$$

with structure constants f^{ABC} .

The fundamental representation matrices: [D]

$$T_{ij}^A, \quad i, j = 1, 2, 3, 4 \quad (123)$$

Normalization: [D]

$$\text{Tr}(T^A T^B) = \frac{1}{2} \delta^{AB} \quad (124)$$

A.2 Branching Rules

Under $SU(4)_C \rightarrow SU(3)_c \times U(1)_{B-L}$: [D]

$$\mathbf{4} \rightarrow \mathbf{3}_{1/3} \oplus \mathbf{1}_{-1} \quad (125)$$

$$\mathbf{6} \rightarrow \mathbf{3}_{-2/3} \oplus \bar{\mathbf{3}}_{2/3} \quad (126)$$

$$\mathbf{15} \rightarrow \mathbf{8}_0 \oplus \mathbf{3}_{-4/3} \oplus \bar{\mathbf{3}}_{4/3} \oplus \mathbf{1}_0 \quad (127)$$

$$\mathbf{10} \rightarrow \mathbf{6}_{2/3} \oplus \mathbf{3}_{-2/3} \oplus \mathbf{1}_{-2} \quad (128)$$

$$\mathbf{20} \rightarrow \mathbf{8}_0 \oplus \mathbf{6}_{4/3} \oplus \bar{\mathbf{6}}_{-4/3} \quad (129)$$

A.3 Clebsch-Gordan Coefficients

For the proton decay operators: [D]

$$\langle \mathbf{1} | \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} \rangle = \epsilon^{\alpha\beta\gamma} \quad (130)$$

The color singlet contraction:

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10} \quad (131)$$

Tensor product decomposition: [D]

$$\mathbf{4} \otimes \bar{\mathbf{4}} = \mathbf{15} \oplus \mathbf{1} \quad (132)$$

B Complete Operator Basis

B.1 Dimension-6 Operators

The complete basis of dimension-6 $\Delta B = 1$ operators: [D]

$$\mathcal{O}_{ijkl}^{(1)} = \epsilon^{\alpha\beta\gamma} \epsilon_{ab} (d_{i\alpha}^c C u_{j\beta}) (Q_{k\gamma}^a L_l^b) \quad (133)$$

$$\mathcal{O}_{ijkl}^{(2)} = \epsilon^{\alpha\beta\gamma} \epsilon_{ab} (d_{i\alpha}^c C Q_{j\beta}^a) (u_{k\gamma}^c L_l^b) \quad (134)$$

$$\mathcal{O}_{ijkl}^{(3)} = \epsilon^{\alpha\beta\gamma} (u_{i\alpha}^c C u_{j\beta}) (d_{k\gamma}^c C e_l) \quad (135)$$

$$\mathcal{O}_{ijkl}^{(4)} = \epsilon^{\alpha\beta\gamma} (u_{i\alpha}^c C d_{j\beta}) (u_{k\gamma}^c C e_l) \quad (136)$$

$$\mathcal{O}_{ijkl}^{(5)} = \epsilon^{\alpha\beta\gamma} (d_{i\alpha}^c C d_{j\beta}) (u_{k\gamma}^c C \nu_l) \quad (137)$$

$$\mathcal{O}_{ijkl}^{(6)} = \epsilon^{\alpha\beta\gamma} \epsilon_{ab} (Q_{i\alpha}^a C Q_{j\beta}^b) (u_{k\gamma}^c C \nu_l) \quad (138)$$

where i, j, k, l are generation indices.

B.2 Fierz Identities

The Fierz identity relates different operator structures: [D]

$$(\bar{\psi}_1 \gamma^\mu P_L \psi_2)(\bar{\psi}_3 \gamma_\mu P_L \psi_4) = -(\bar{\psi}_1 \gamma^\mu P_L \psi_4)(\bar{\psi}_3 \gamma_\mu P_L \psi_2) \quad (139)$$

This allows reduction to a minimal basis.

Additional Fierz relations: [D]

$$(\bar{\psi}_1 \psi_2)(\bar{\psi}_3 \psi_4) = -\frac{1}{4}(\bar{\psi}_1 \gamma^\mu \psi_4)(\bar{\psi}_3 \gamma_\mu \psi_2) + \dots \quad (140)$$

$$(\bar{\psi}_1 \sigma^{\mu\nu} \psi_2)(\bar{\psi}_3 \sigma_{\mu\nu} \psi_4) = \dots \quad (141)$$

C Phase Space Calculation

C.1 Two-Body Decay Kinematics

For $p \rightarrow e^+ \pi^0$ in the proton rest frame: [D]

$$E_e = \frac{m_p^2 + m_e^2 - m_\pi^2}{2m_p} \quad (142)$$

$$E_\pi = \frac{m_p^2 + m_\pi^2 - m_e^2}{2m_p} \quad (143)$$

The common momentum magnitude (neglecting m_e):

$$|\vec{p}| = \frac{\lambda^{1/2}(m_p^2, m_e^2, m_\pi^2)}{2m_p} \approx \frac{m_p^2 - m_\pi^2}{2m_p} \quad (144)$$

where $\lambda(a, b, c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$.

The Källén function: [D]

$$\lambda(a, b, c) = (a - b - c)^2 - 4bc \quad (145)$$

C.2 Phase Space Factor

The kinematic factor in the width formula: [D]

$$\mathcal{K}(m_p, m_\pi) = \frac{m_p^3}{(m_p^2 - m_\pi^2)^2} \quad (146)$$

For the inverse (in lifetime formula):

$$\mathcal{K}^{-1} = \frac{(m_p^2 - m_\pi^2)^2}{m_p^3} \quad (147)$$

Two-body phase space integral: [D]

$$\int d\Phi_2 = \frac{1}{8\pi} \frac{|\vec{p}|}{m_p} \quad (148)$$

D Dimensional Analysis

D.1 Dimension Counting

Quantity	Mass Dimension	Units
$\mathcal{O}_6$	6	$[\text{mass}]^6$
M_X	1	$[\text{mass}]$
g_{PS}	0	dimensionless
α_H	3	$[\text{mass}]^3$
Γ	1	$[\text{mass}] = [\text{time}]^{-1}$
τ_p	-1	$[\text{time}]$
C_6	-2	$[\text{mass}]^{-2}$

D.2 Dimensional Check

For the width formula: [\[D\]](#)

$$[\Gamma] = \frac{[g]^4}{[M_X]^4} [C]^2 [\alpha_H]^2 \frac{[m_p]^4}{[m_p]^3} = \frac{1 \cdot 1 \cdot [\text{mass}]^6 \cdot [\text{mass}]}{[\text{mass}]^4} = [\text{mass}]^1 \quad (149)$$

confirming dimensional consistency. ✓

For the operator coefficient: [\[D\]](#)

$$[C_6] = \frac{[g]^2}{[M_X]^2} = [\text{mass}]^{-2} \quad \checkmark \quad (150)$$

For the matrix element: [\[D\]](#)

$$[\alpha_H] = [\langle \pi | \mathcal{O}_6 | p \rangle] = [\text{mass}]^3 \quad \checkmark \quad (151)$$

E Formula Catalog

E.1 Core Formulas

1. **PS Group:** [\(1\)](#)
2. **Generator Normalization:** [\(10\)](#)
3. **Hypercharge:** [\(24\)](#)
4. **Boson Decomposition:** [\(40\)](#)
5. **X Charges:** [\(42\)](#)
6. **X Mass:** [\(50\)](#)
7. **Dim-6 Operator:** [\(55\)](#)
8. **Amplitude:** [\(78\)](#)
9. **Partial Width:** [\(82\)](#)
10. **Lifetime:** [\(91\)](#)
11. **API-PD1:** [\(104\)](#)
12. **No-Backflow:** [\(114\)](#)

E.2 Invariance Checks

The formulas satisfy: [D]

1. Lorentz invariance: All tensors properly contracted
2. Gauge invariance: Color singlet final states
3. Dimensional consistency: All equations balanced
4. Hermiticity: Width formula is real and positive

E.3 Consistency Relations

Scaling consistency: [D]

$$\tau_p \cdot \Gamma = 1 \quad \checkmark \quad (152)$$

Dimensional consistency: [D]

$$[\tau_p] = [\Gamma]^{-1} = [\text{mass}]^{-1} \quad \checkmark \quad (153)$$

F Renormalization Group Aspects

F.1 Operator Mixing

The dimension-6 operators mix under renormalization: [D]

$$\mu \frac{d}{d\mu} C_i = \gamma_{ij} C_j \quad (154)$$

where γ_{ij} is the anomalous dimension matrix.

Leading-order anomalous dimensions: [D]

$$\gamma = \frac{\alpha_s}{4\pi} \gamma^{(0)} + O(\alpha_s^2) \quad (155)$$

F.2 Running Effects

The Wilson coefficient at low scale: [D]

$$C(\mu_{\text{low}}) = \left(\frac{\alpha_s(M_X)}{\alpha_s(\mu_{\text{low}})} \right)^{d/b_0} C(M_X) \quad (156)$$

where d is the anomalous dimension coefficient. [Dc]

The enhancement factor: [D]

$$A = \left(\frac{\alpha_s(M_X)}{\alpha_s(m_p)} \right)^{d/b_0} \quad (157)$$

Document Hash: v61-[computed at release]
Status: PROGRAM NOTE — OPEN until M_X derived
Layer A: Structural derivation complete
Layer B: Comparison framework defined (quarantined)